EUGLOH Annual Summit 2026

Science and the City: Exploring Pathways for Global Health Innovation
University of Hamburg

WS4: From the **European Student Card Vision** to Real Service Access via **eduTAP**

Alexander Loechel (LMU Munich) &
Morgan Persson (Lunds Universitet)

15. September 2026 – WS4

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Agenda

Time	Session	Whom	Goal
11:30 – 11:40	Welcome and Understanding the Participants	Alexander Loechel Morgan Persson	Shared view of backgrounds and priority use cases
11:40 – 11:50	From the European Student Card Vision to eduTAP	Alexander Loechel	Conceptual foundation: ESC gap and the Triple-A model
11:50 – 12:00	Institutional Use Cases	Morgan Persson	One candidate use case per group
12:00 – 12:30	Designing an eduTAP Deployment	Alexander Loechel	High-level institutional deployment blueprint
12:30 – 12:40	Pass Design	Alexander Loechel	Minimum pass content for the selected service
12:40 – 12:50	Reader and Service Configuration	Morgan Persson	End-to-end demo: issue, present, validate, authorise
12:50 – 13:00	Conclusions, Next Steps, and Questions	Alexander Loechel Morgan Persson	Concrete first step towards an eduTAP pilot

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This is a workshop – not another presentation

> *Talk is cheap. **Show me the code.***
>
> — Linus Torvalds, linux-kernel mailing list, 2000



60 of the next 90 minutes are yours: your services, your deployment, your reader configuration.

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Welcome and Understanding the Participants

Part 1 · 11:30 – 11:40 · Alexander Loechel & Morgan Persson

Where do you start from – and which service do you want to unlock?

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Learning Objectives

By the end of the workshop, participants will be able to …

- explain how eduTAP complements and **operationalises** the **European Student Card vision**;
- distinguish between **identification, authentication, authorisation**, and accounting;
- identify relevant eduTAP **use cases** for their own institution;
- understand the main roles, components, and integration points of an eduTAP deployment;
- compare on-premises and managed-cloud deployment scenarios;
- understand the basic process of designing a service pass and configuring a reader;
- identify the stakeholders and next steps required for a first institutional pilot.

Who Is in the Room?

1. Which institution and organisational area do you represent?
2. Which group you represent:
   students, administrative staff, faculty staff, service provider or external partner
3. Do you work primarily in research, IT, student services, mobility, libraries, campus management – or another area?
4. Does your institution already use physical campus cards, mobile credentials, or wallet passes?
5. Does your institution use eduGAIN and Shibboleth
6. What is your favourite service while on mobility at a host university

Also: your expectations and current challenges.

Output: a shared view of the participants' starting points and priority use cases.

The Guiding Question

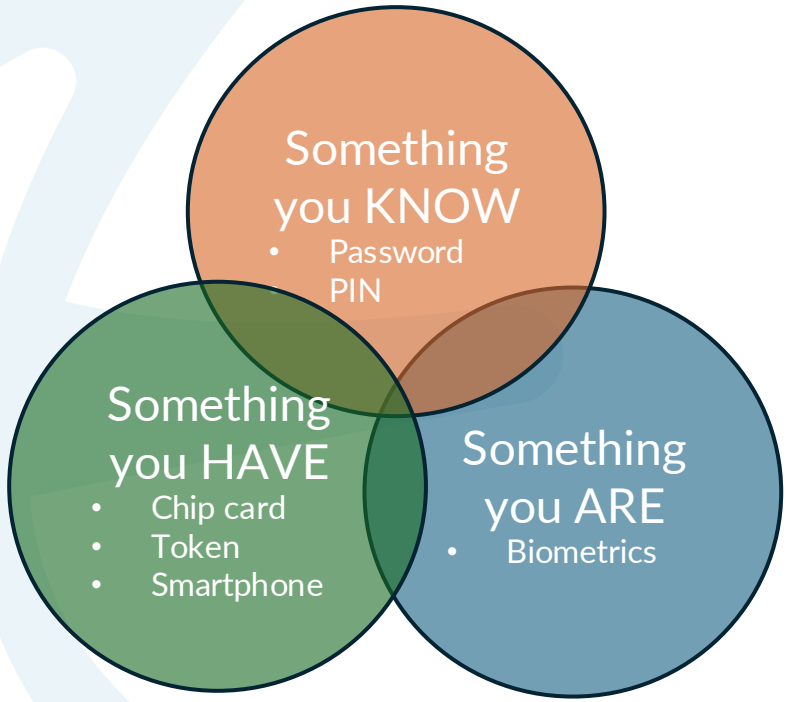
How can a verified university identity
(by the home university)
be translated into secure and convenient
access to a real service?

Identity → Credential → Authentication → Authorisation → Accounting → Service

Key to all Services → Identification / Authentication

- **Triple-A-System**
  - (Identification)
  - **Authentication**
  - **Authorization**
  - **Accounting**

Authentication Factors



User Experience is key – Benchmarks

Providing best of class user experience is about seamless and clear **service access**

- For Member of the Higher Education Institution
- For members of other Higher Education Institutions / Partners
- For incoming members of other Higher Education Institutions / Partners → short- & long-term Mobility

For the Person **User Experience** is key, it should be as easy and simple as possible

- Benchmark Examples

WLAN Access

Login to Online Services

→ To access services (physical) on site it should be as easy as using the Wifi with eduroam

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From the European Student Card Vision to eduTAP

Part 2 · 11:40 – 11:50 · Alexander Loechel

From recognised identity to real service access – the Triple-A model

It Is About Accessing Services

Identification
- **Identification**
- **Status verification**
- **Proof of entitlement**
- Single Sign On / 2FA / FIDO2
- Attendance check
  - *Check-in for exams*
  - *attendance at courses*
  - *Time recording*
- Electronic Signature (*of legal documents*)

Electronic payment / cashless campus
- Canteen & cafeterias
- Vending machines
- Printing / scanning (*secure & follow me printing*)
- Ticketing (*events and conferences*)

Physical Access Control
- Areas (*campus, parking lot*)
- Buildings
- Rooms
  - Classrooms
  - Labs
  - Computer rooms
  - Learning spaces
  - Offices
  - Accommodation facilities *(i.e., dorms)*
- Athletic facilities

Library services
- Access to / borrow
  - Physical media (book, audio and video media)
  - E-media (book, audio and video media)
- Learning spaces

Transport
- On campus services (*university shuttle service*)
- Public transport tickets & discounts

Discount and promotions
- **Discounts** on cultural activities
  - Museums
  - theaters
  - cinemas
- Shops
- Restaurants

On- & Off-Campus → On-Site Usage / Proximity Use-Cases + Online Services





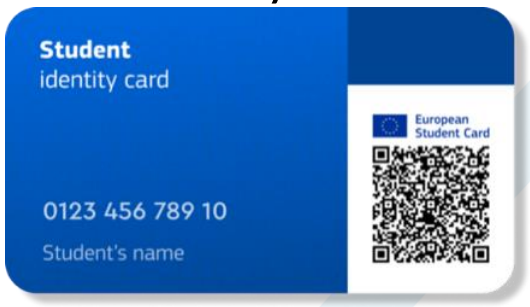


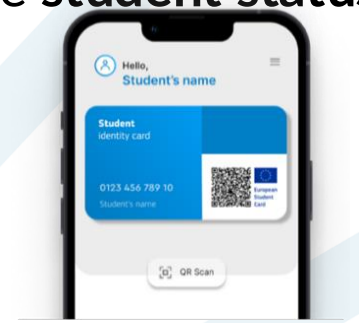


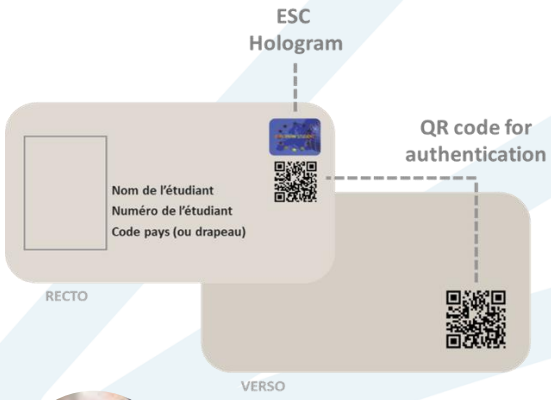
The European Student Card (ESC)

The **European Student Card** – a project to add a **European dimension** to local Student IDs

✓ Securely **validate** the **student status**









Idea by →
Jean-Paul Roumegas
(les Crouse)

European Student Card logo

Combines the European identity (text & flag) with the QR code. It facilitates the implementation and recognition of the ESCs

ESCN - European Student Card Number

Unique card identifier that serves to link a specific card to the specific higher education institution who produced it and to identify a card

ESI - European Student Identifier

Unique identifier that allows to link a given student to a specific ESC. It is not recommended to be printed on the card.

✓ A great vision

European Student Card





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The European Student Card (ESC)

Benefits **of the European Student Card Initiative**

- **For students:**
  - → Easy access to course materials prior to mobility, online course registration and automatic recognition of ECTS credits
  - → **Immediate access to services at the host university, such as libraries, transport and accommodation**
  - → **Discounts on cultural activities throughout the EU**
- **For Higher Education Institutions:**
  - → Easy, online management of the entire mobility process – from student selection to the recognition of ECTS credits
  - → **Online identification of students**, simplified and secure exchange of student data - including academic records - between Higher Education Institutions
  - → Reduced administrative burden associated with student mobility

Source: https://ec.europa.eu/education/education-in-the-eu/european-student-card-initiative_en

→ **European Student Card Project** as implementing instance

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The European Student Card (ESC)

Basic concepts

The European Student Card (ESC) is a new standard created by European Higher Education Institution (HEI) **to promote student mobility within Europe** (EHEA zone).

→ The long-term objective is **to give all European Students the same rights as a local student** when travelling abroad.

So, the idea is to **create a card that can be accepted anywhere within Europe**. This card **gives access to an unlimited number of services** that students use in their daily life, such as borrowing book from university library, accessing student cafeteria with discounted prices, accessing university premises, using public transportation etc.

Short- and long-term perspectives

The first basic service to be set up is **student status authentication**. This service offers the possibility for any organization (private business, public administration, association, HEI...) to confirm whether a card holder is a student or not. **This service is a necessity because student status authentication is usually a prerequisite to give access to other services that are restricted to students**.

- **Other services will become available afterward**, such as:
  - The possibility to use the card as an **electronic wallet** for day-to-day expenses
  - The possibility to write new information on the card after it has been delivered, granting access to a wide range of new services for the student
  - The possibility of using the card as a **key** to allow **exchange of data between HEIs**, to facilitate administrative procedures during student mobility
- **Dematerialisation of the card itself**, using a smartphone

Source: ESC Handbook for institution: Users's Guide to connect to the Platform (http://europeanstudentcard.eu/wp-content/uploads/2017/02/ESC-User-Guide-def.pdf)

→ Interoperability

The European Student Card (ESC)

European Student Card

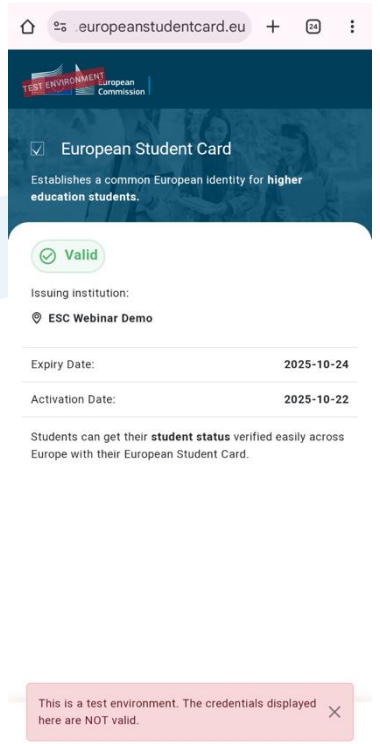






Seamless access to physical and digital services
while on mobility, e.g. Library, Public Transport, Physical Access







One Credential to rule them all?

With a QR-Code or a verifiable credential?

→ not realistic

Gall's Law

A complex system that works is invariably found to have evolved from a simple system that worked. The inverse proposition also appears to be true:
*A **complex system designed from scratch never works and cannot be made to work**.*
You have to start over, beginning with a working simple system.

— John Gall, “Systemantics: How Systems Really Work and How They Fail”, 1975

 The European Student Card Vision can not be solved top-down with the current technical specification and implementation

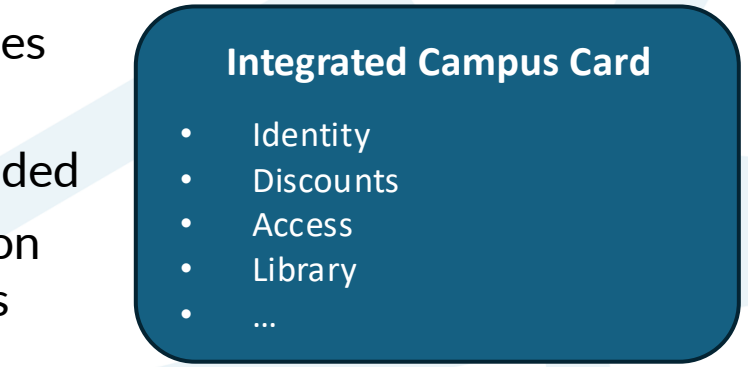
 A new approach is required, based on existing, proven technologies, and a simple working system

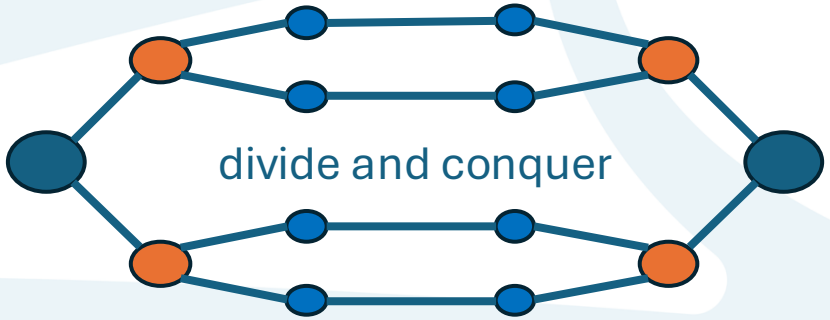
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Campus Cards from a User Experience Point of View

User preferences for **Smartphone Wallets:**

- NFC / Contactless passes are better than scanning visual codes
- **One pass per service** is better
  - User control over which services to use
  - Auto-Presentment of pass needed
  - Clear contextual communication (additional information on pass specific for the service, specific branding of service)



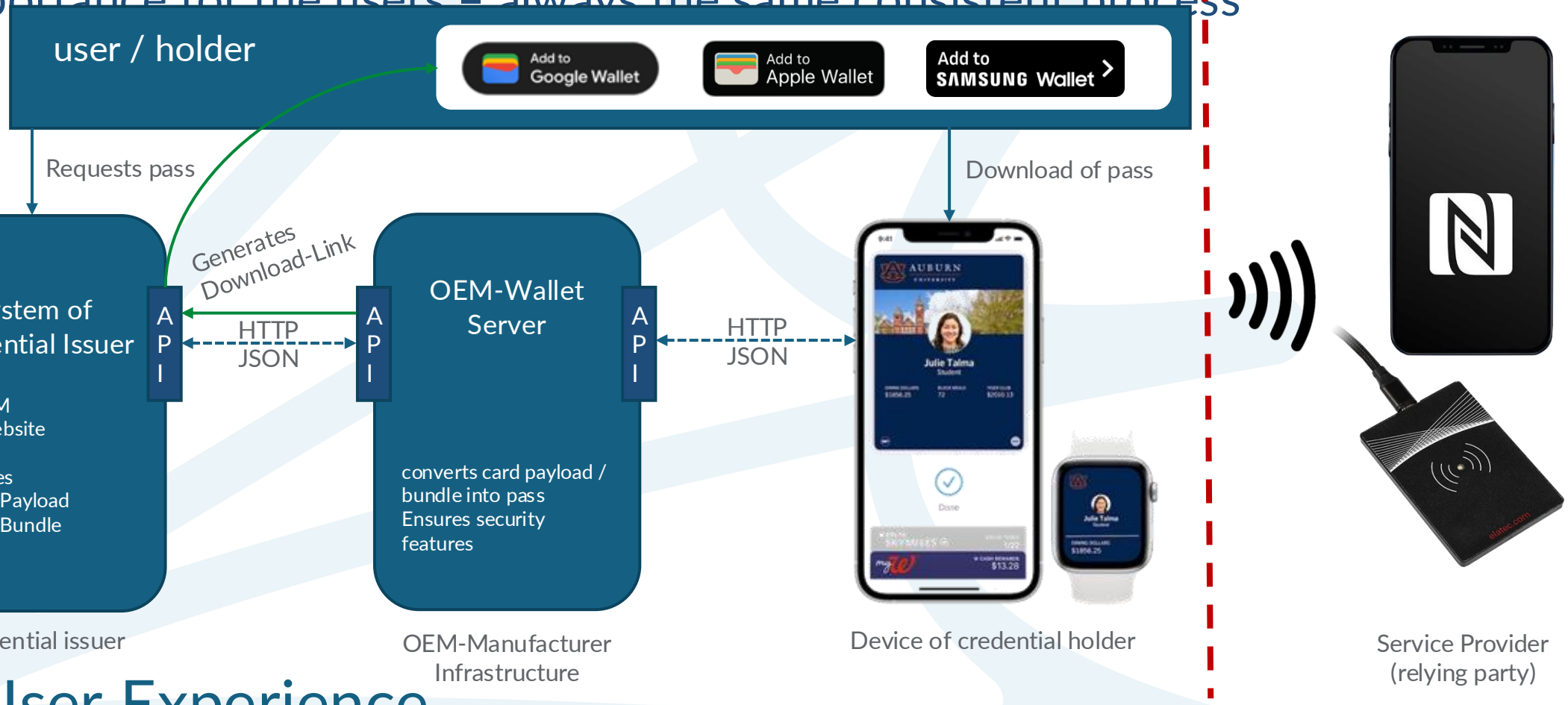




Issuing process of wallet passes reedeming

→ importance for the users – always the same consistent process

- **Web-Provisioning**
- App-Provisioning
- Wallet-Provisioning
- *Push-Provisioning*



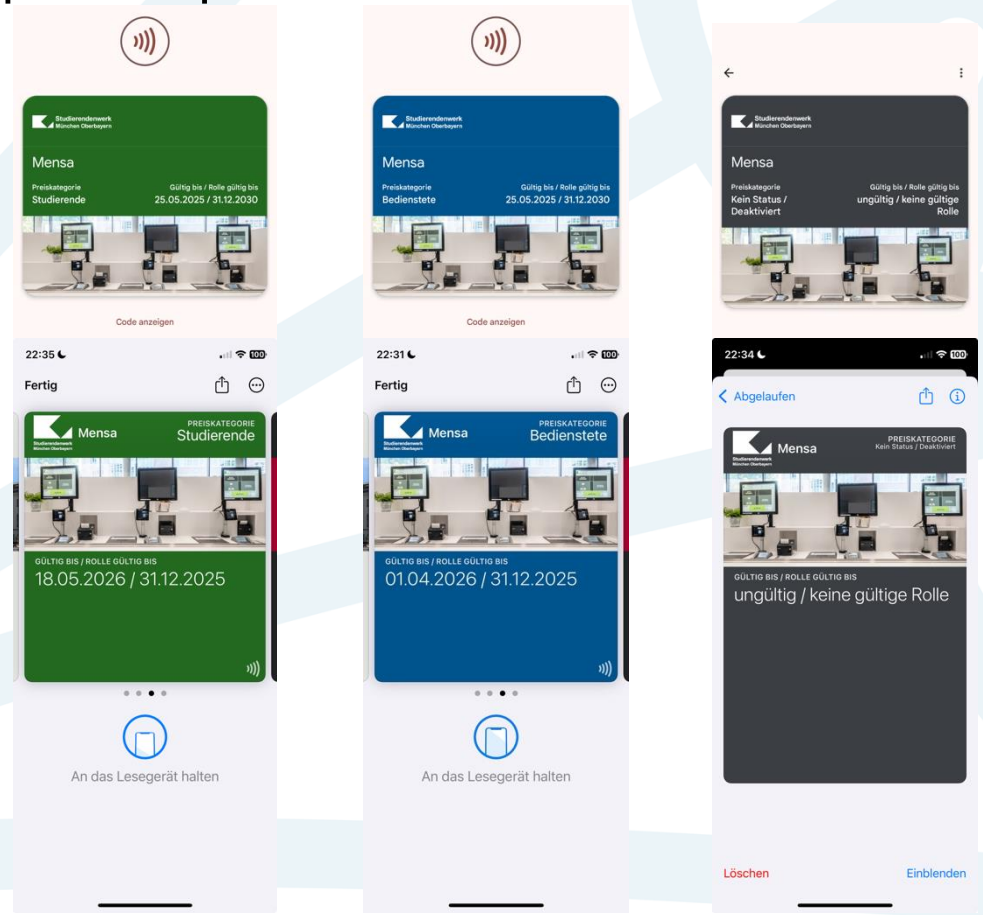
→ User Experience

Discount-Pass for Mensa Studierendenwerk München Oberbayern

Eligibility verification for discounts with the Apple / Google Wallet via NFC, payment with an Open-Loop Bank / Credit-Card

Google Wallet
Since Feb. 2025

Apple Wallet



Discount-Pass for Mensa Studierendenwerk München Oberbayern

Payment Process in the Mensa

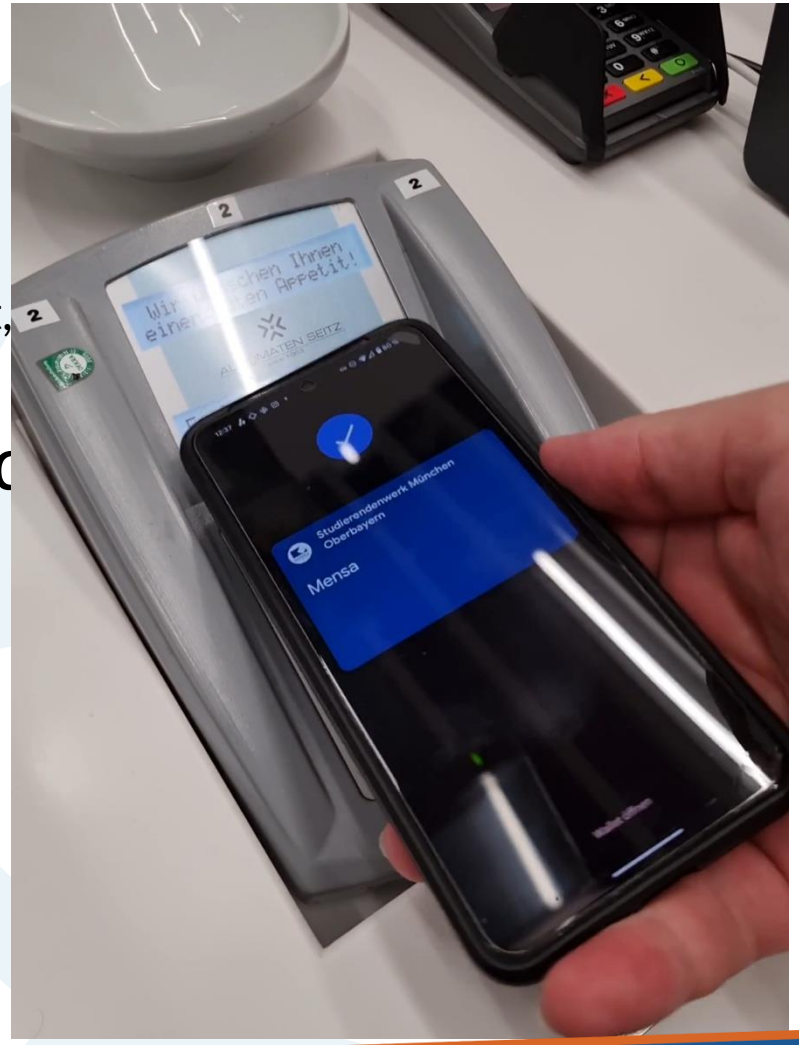
Double Tap:

1. Eligibility verification of status group (Student,
   → Discount will be applied
2. Pay with an Open- ark ard









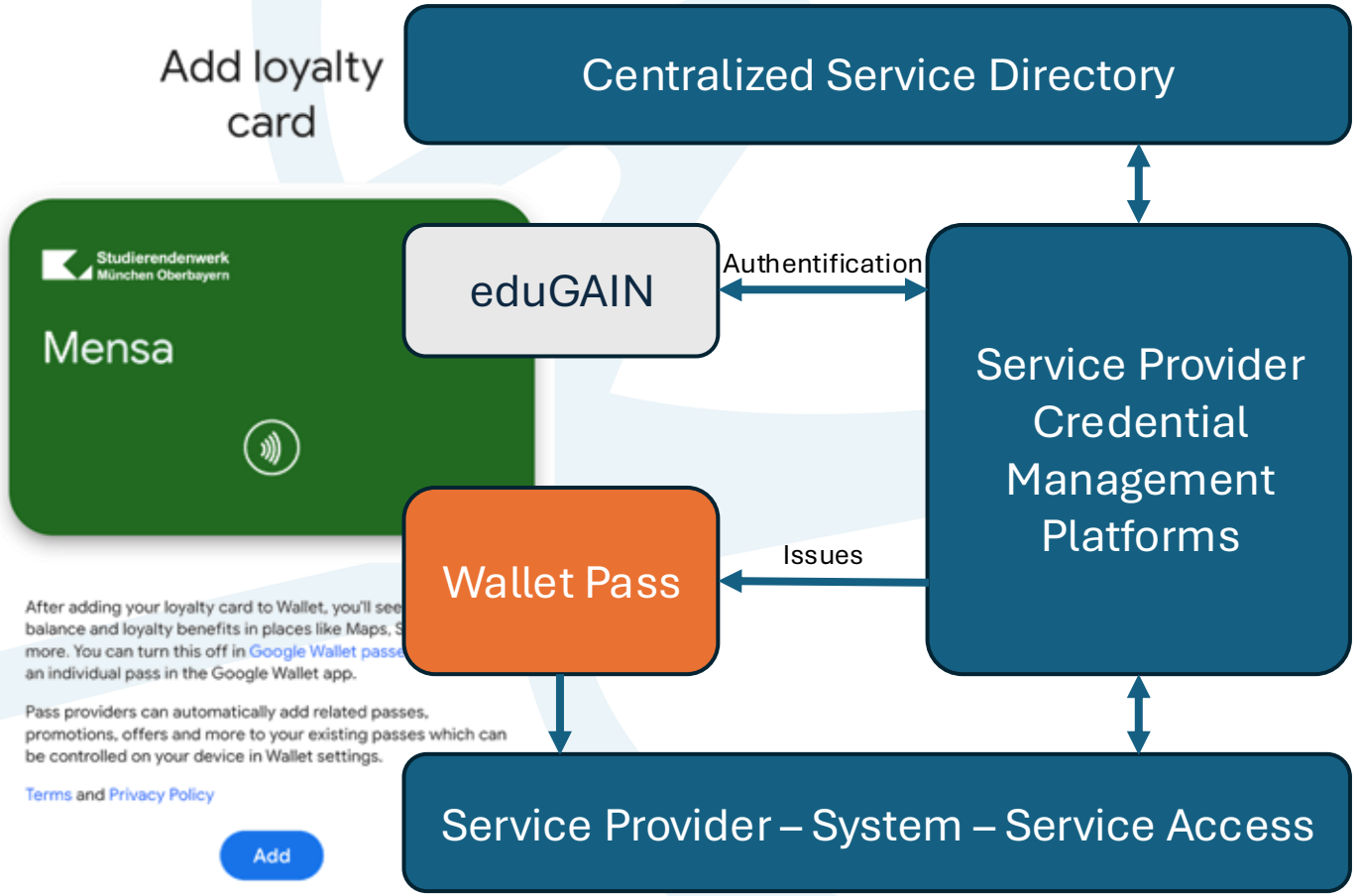
Transition of credentials into the Smartphone Wallets
Dedicated Pass Portals for login via eduGAIN



https://mensapass.stwm.de/

Currently only open to Munich HEIs, as guests don‘t get discounts, but can pay directly via Open-Loop-Payment

→ *Discounts for Alliance Members are politically discussed and in preparation*



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eduTAP – educational tapping



tap to …

- identify yourself,
- verify your status,
- proof your entitlement,
- claim a discount,

tap to access a service on-site directly:

- tap to pay (open and close loop-payments),
- tap to open a door,
- tap to lend a book,
- tap to take a campus shuttle.

*no app, **just tap**!*

https://eduTAP.eu

eduTAP in one picture

From a verified university identity to real service access – one framework, any medium, one tap.

1 · Verified identity

- Home university IdM / AAI
- eduGAIN attributes: who is a student, staff, alumni?
- Trusted source, no new registry

2 · Credential management

- eduTAP CMP: issuer & pass portals
- Login via eduGAIN – works across HEIs
- Lifecycle: issue, update, revoke, notify

3 · Credential, any medium

- One identity pass (ISO/IEC 18013-5)
- Separate passes per service
- Wallet, card, QR or verifiable credential

4 · Service access by tap

- Reader / verifier at the service
- Authentication → Authorisation → Accounting
- Library, door, canteen, shuttle, discounts

eduTAP is a framework, not a wallet and not an app – the wallet is just one medium.

Open standards
ISO 14443I, SO/IEC 18013-5, eduGAIN (SAML, OIDC)

Reusable components
CMP, issuer, portals, verifiers

Deployment blueprint
Containers, roles, layers, integration

Community
HEIs, alliances, service providers

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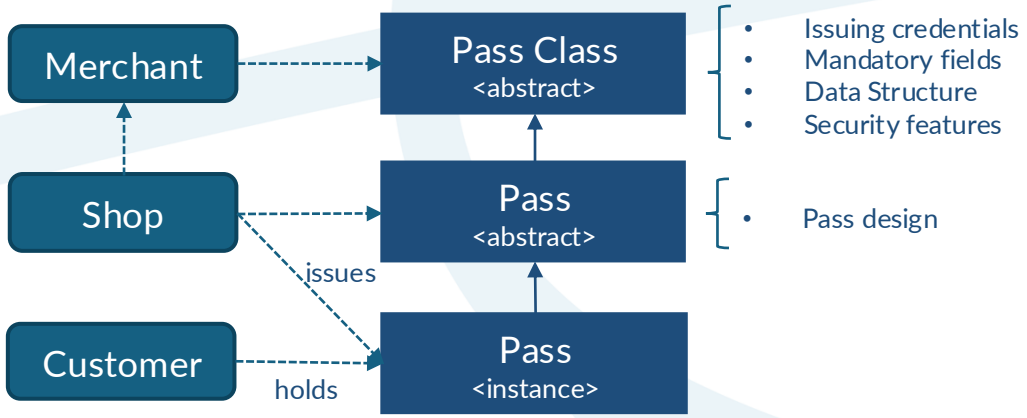
Technical details and features of Wallets / Passes

The wallet is a **container for multiple passes**

Technical features:

- **Auto-Presentment / Smart Select ← reader selects the specific pass to interact with** (if available)
- Selectable Authentication requirements ← **security features**
- Express-mode (Apple)
- Battery reserve (pass is still usable if phone has powered off; important for keys, transit tickets, payment)
- Groupable
- Validity dates
- Remote updateable / revokable
- Offline usable (→ non traceable)

- *Host Card Emulation possible*



Transition of credentials into the Smartphone Wallets
Examples: Active and potential IDs for LMU Munich

Google Wallet

Apple Wallet



→ all **NFC enabled** → **NFC enabled** where possible

Add to Google Wallet

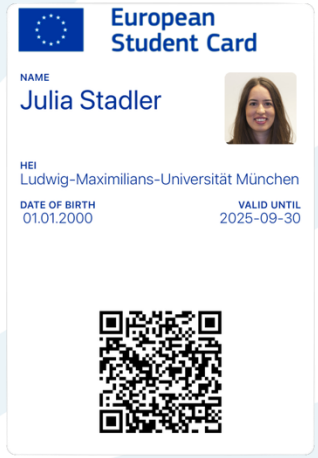
Add to Apple Wallet

Transition of credentials into the Smartphone Wallets
The European Student Card

The European Student Card (v1.1 new ESC Design)

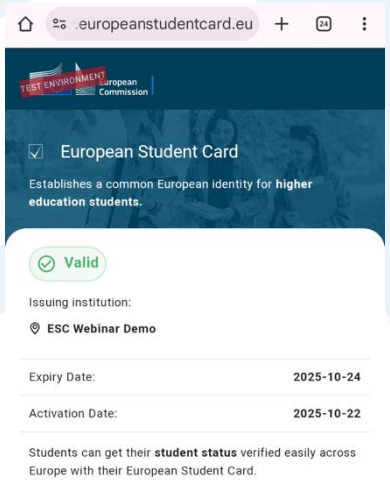


Google



Apple

- ✓ Discounts off-campus
- ✓ European dimension → Erasmus+ marketing
- ✓ Online status verification



Transition of credentials into the Smartphone Wallets

Library ID

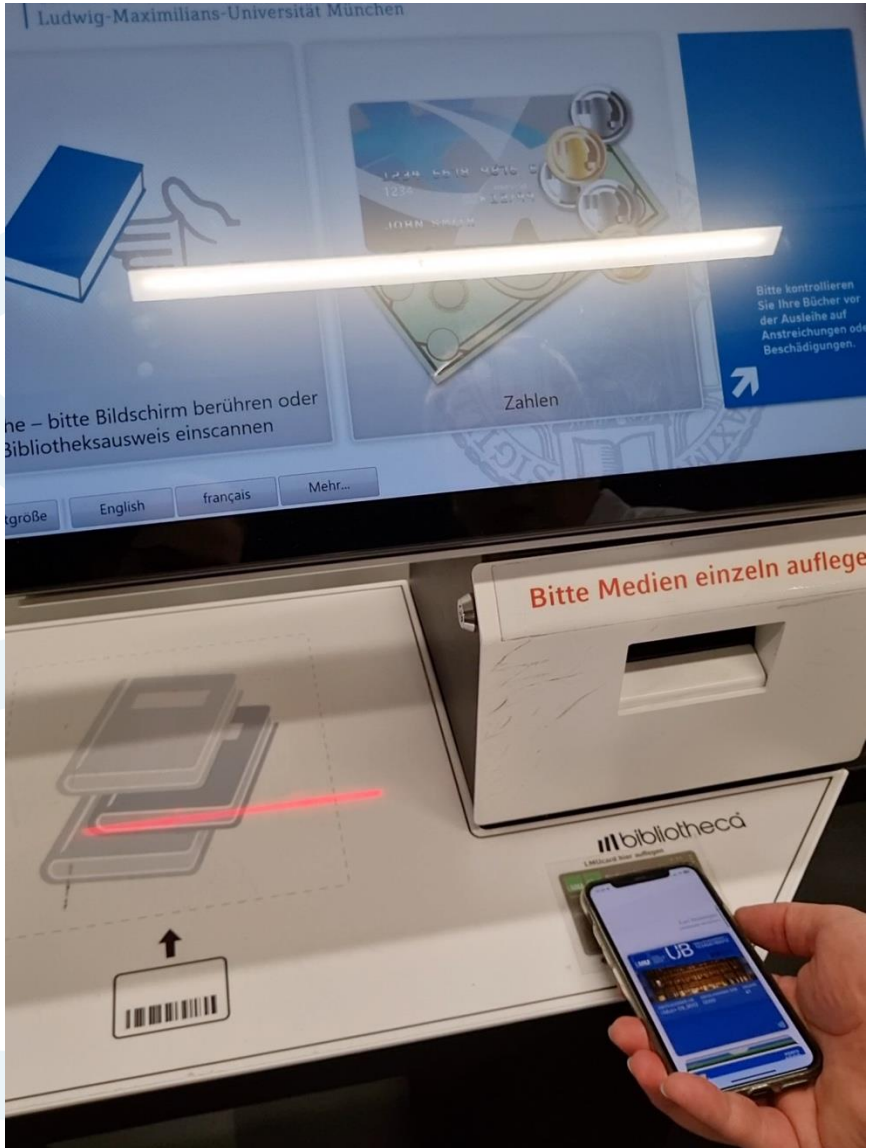
NFC enabled by default



Google



Apple

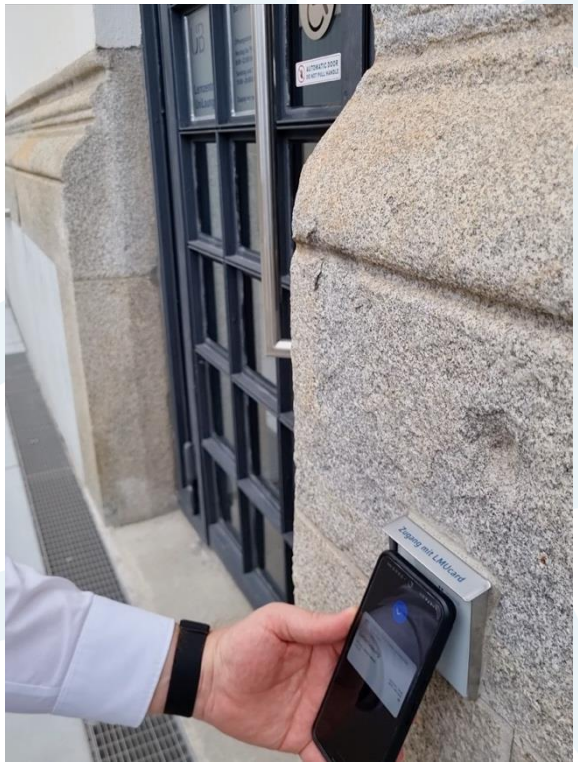


Transition of credentials into the Smartphone Wallets
Use Case: Physical Access Control (Access to learning spaces)

Physical access to the learning spaces (provided by the library)



LMUcard



Google Smart Tap

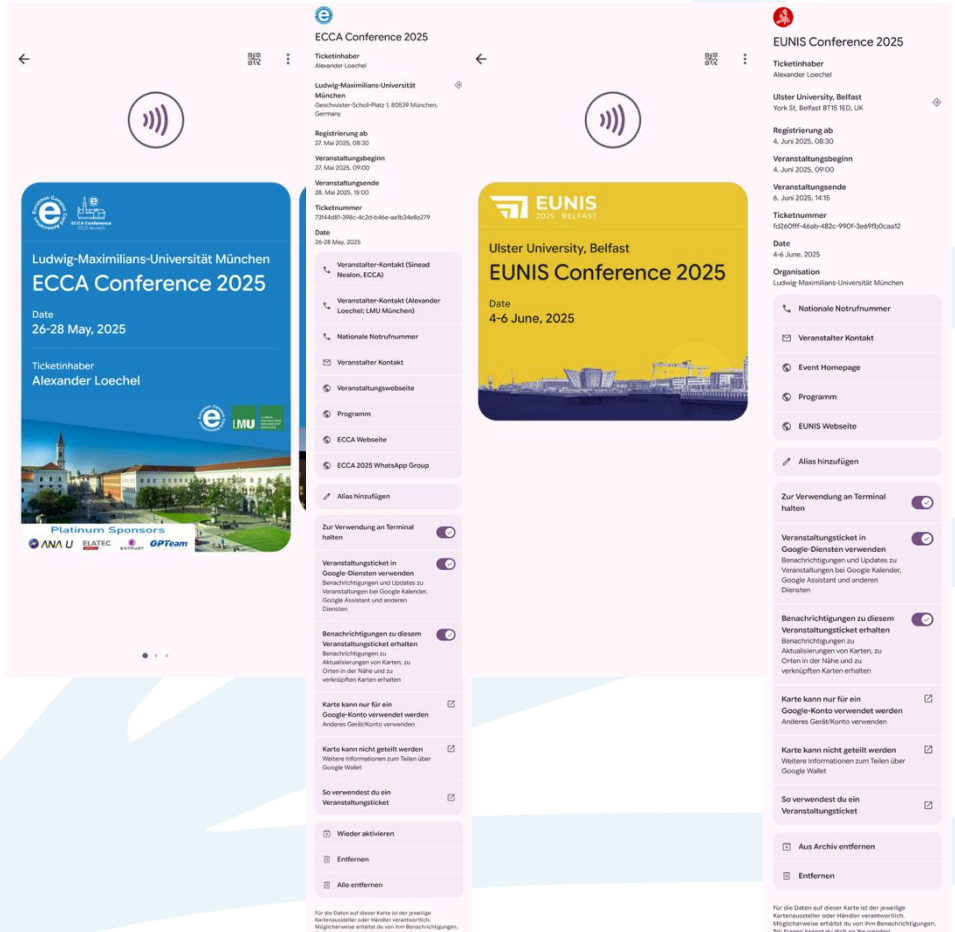


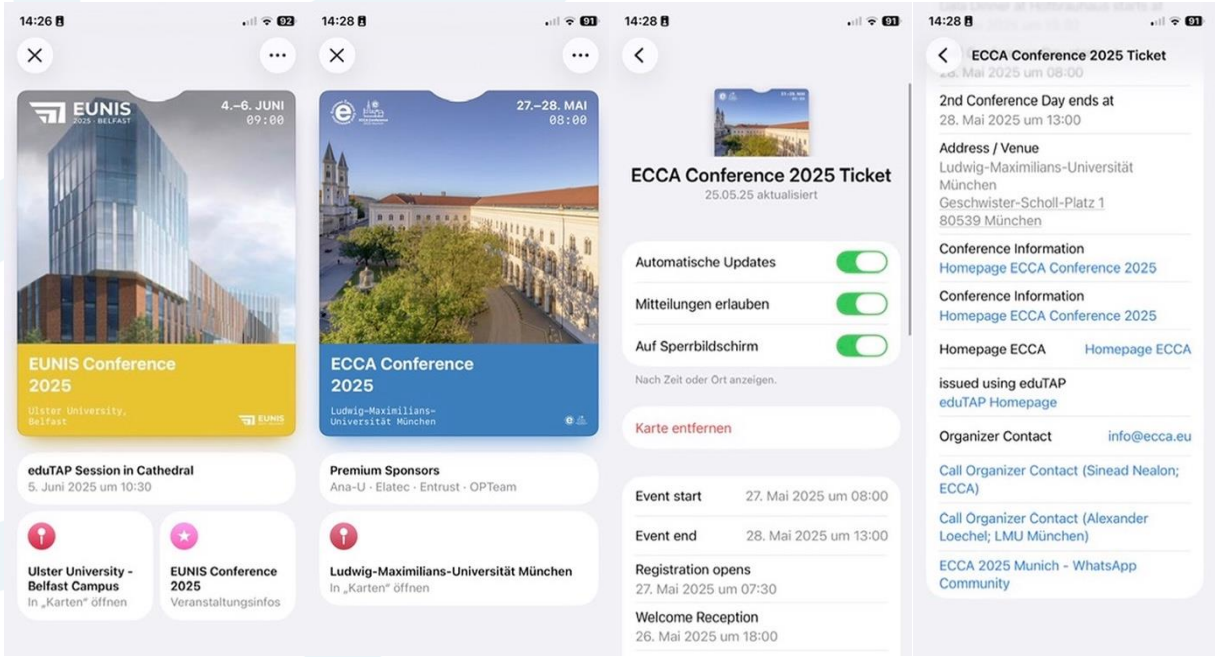
Apple

Access

Mobile Identity → Credentials
Communicate with holders

- Wallet passes are updateable, revokable, and have communication channels
- The also have interactive elements



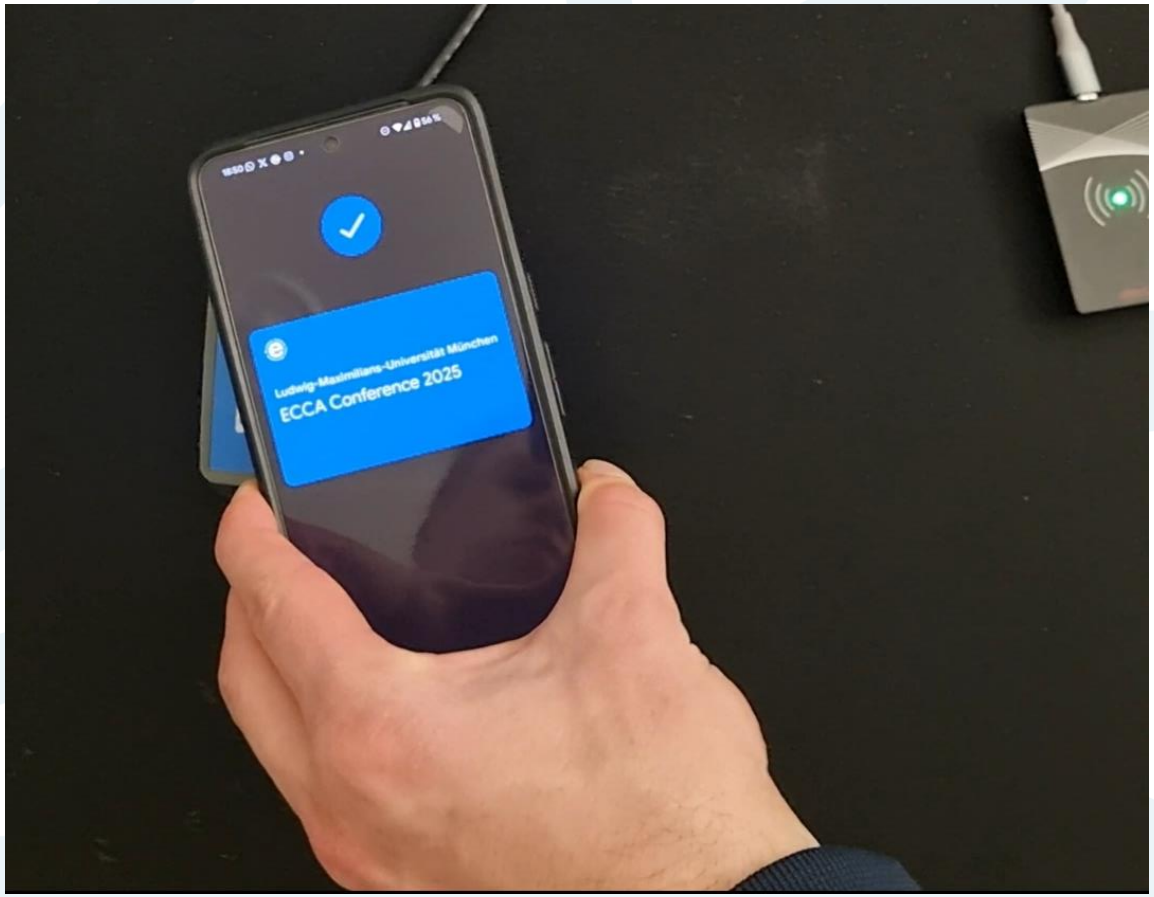


- Ability to send messages to a pass
- Pop-Up on time- or location bases events
- Geo-fencing of passes

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Technical details and features of Wallets / Passes

Auto-Presentment / Smart Select of an NFC enabled Pass in the Wallet



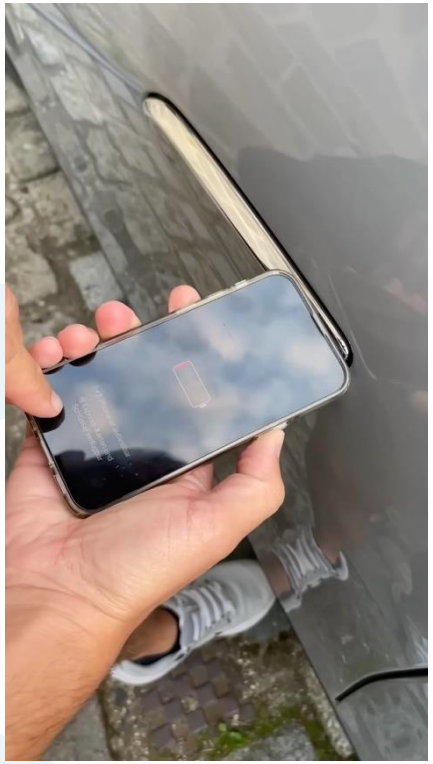
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Technical details and features of Wallets / Passes

Battery Reserve Mode – Use the Wallet even if it‘s powered of

- for Keys (Dorm-Rooms, Class-Rooms, Building Access, Cars)
- for Payment
- for Transport Tickets





Guaranteed to work 5 hours, to
up to 7 days after shutdown of the smartphone

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Service Domains: Identity and Discount

From a legal point of view, Identity Documents of public universities are **official state photo IDs**

- ✓ Legal requirements
- ✓ Security considerations
- ✓ High assurance level

→ **International standards** for HEI Identities → eduGAIN (incl. eduPerson, SCHAC)
e.g. European Student Identifier → routable information for course participation at other institutions (during mobility programmes)

Identity Documents (Student ID, Staff ID and Affiliate IDs) are used

On-Site (proximity use-cases)

On-Campus

- ✓ Attendance at courses
- ✓ Check-in for exams
- ✓ Time recording
- ✓ Discounts in canteen & cafeterias
- ✓ Identification for university authorities (Student legal office and examination offices, campus security)

Off-Campus

- ✓ Identification for public authorities (e.g., police)
- ✓ Identification for third party (e.g., checkin at hostels)
- ✓ Eligibility verification for discounts and access (e.g., role-based, age-based, institution related)

Online

- ✓ As authentification factor

Verifiable Credential for HEIs (Prototypes)

European Student Card as Verifiable Credential



DC4EU non-foundational Ids

Table with available schemes for non-foundational IDs

Scope	Data model name	Brief explanation	Status/ Detailed explanation	Schema URL	Registry URL
Non-foundational identity	EducationalID	Identifies the natural person in the context of an educational organisation, including national extensions	Available	Schema	Verifiable Data Registry
Non-foundational identity	AllianceID	Identifies a student or staff member as affiliated with a European university alliance	Available	Schema	Verifiable Data Registry
Non-foundational identity	EuropeanStudentCard	European Student Card for student mobility, based on DG-EAC's service	Available	Schema	Verifiable Data Registry
Non-foundational identity	MyAcademicID	Identity credential for student mobility, based on MyAcademicID and eduGAIN infrastructure	Available	Schema	Verifiable Data Registry
Non-foundational identity	ProfessionalID	Identity credential for , based on	Available	Schema	Verifiable Data Registry
Non-foundational identity	DoctorID	Identity credential for , based on	Available	Schema	Verifiable Data Registry
Non-foundational identity	EngineerID	Identity credential for , based on	Available	Schema	Verifiable Data Registry

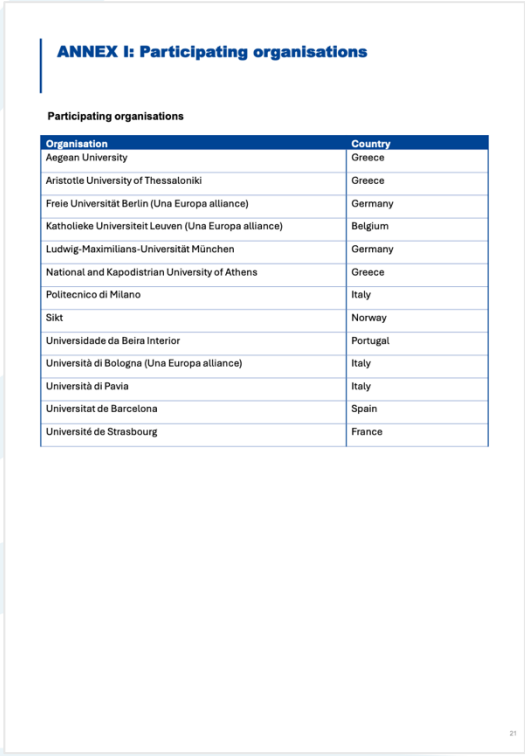
https://github.com/dc4eu/educational-pilot/tree/main/sectorial-eaa-catalogue

Mobile Identity → Digital Identity
European Student Card Verifiable Credential Pilot Report

- W3C Verifiable Credential Data Model v1
- ELM (European Learning Model) Based Data Model – Attribute Structure

→ Revocation / Update Process required

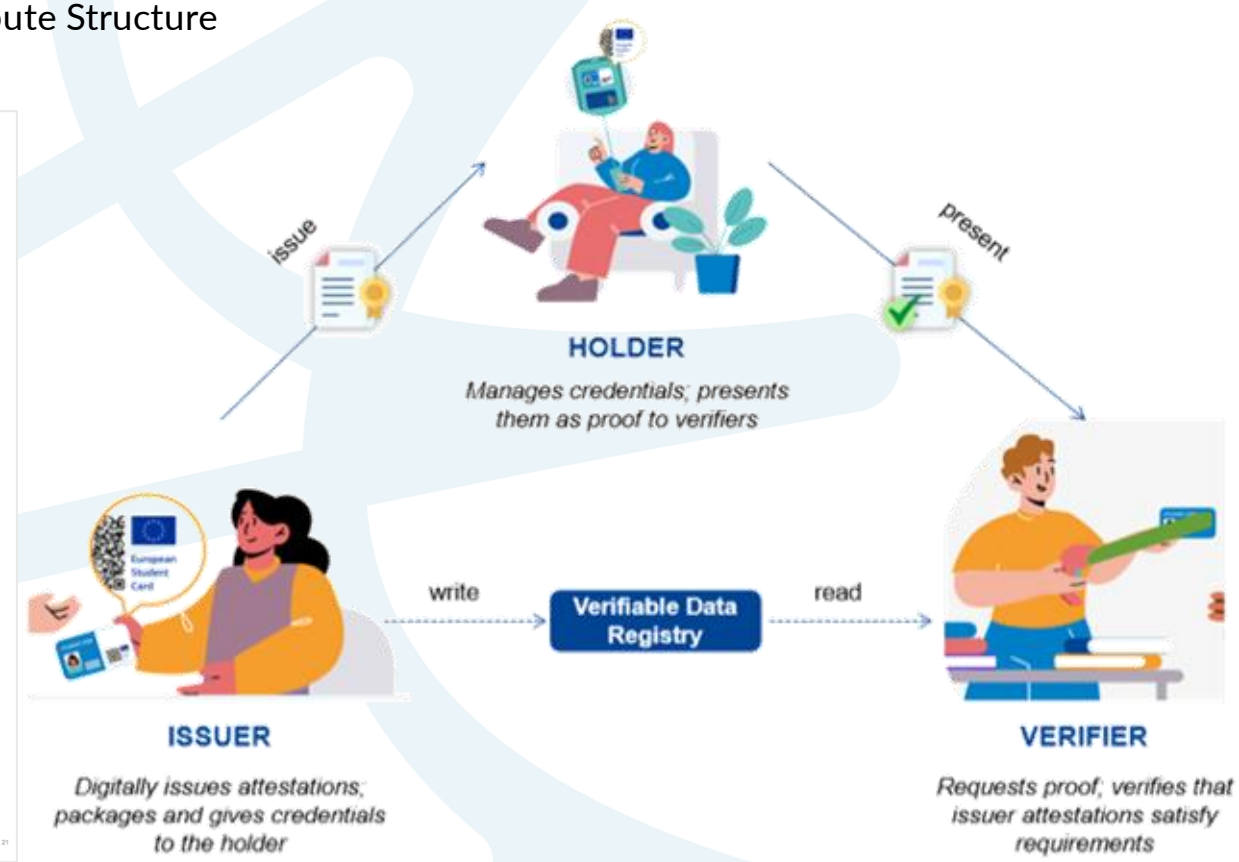




ANNEX I: Participating organisations

Participating organisations

Organisation	Country
Aegean University	Greece
Aristotle University of Thessaloniki	Greece
Freie Universität Berlin (Una Europa alliance)	Germany
Katholieke Universiteit Leuven (Una Europa alliance)	Belgium
Ludwig-Maximilians-Universität München	Germany
National and Kapodistrian University of Athens	Greece
Politecnico di Milano	Italy
Sikt	Norway
Universidade da Beira Interior	Portugal
Università di Bologna (Una Europa alliance)	Italy
Università di Pavia	Italy
Universitat de Barcelona	Spain
Université de Strasbourg	France



https://erasmus-plus.ec.europa.eu/news/piloting-a-digital-european-student-card-as-a-verifiable-credential

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Digital Identity in the Wallet

Common Verifiable Credential Formats / Protocols for Digital Identities

- SD JWT (EU)
- ISO/IEC 18013-5 (mDL & mDoc)
- W3C Verifiable Credential Data Model

Issuing and Redemption Protocols

- OID4VC (OID4VCI and OID4VP)
- ISO/IEC 18013-5

For **proximity** use-cases **NFC / BLE transmission** is crucial

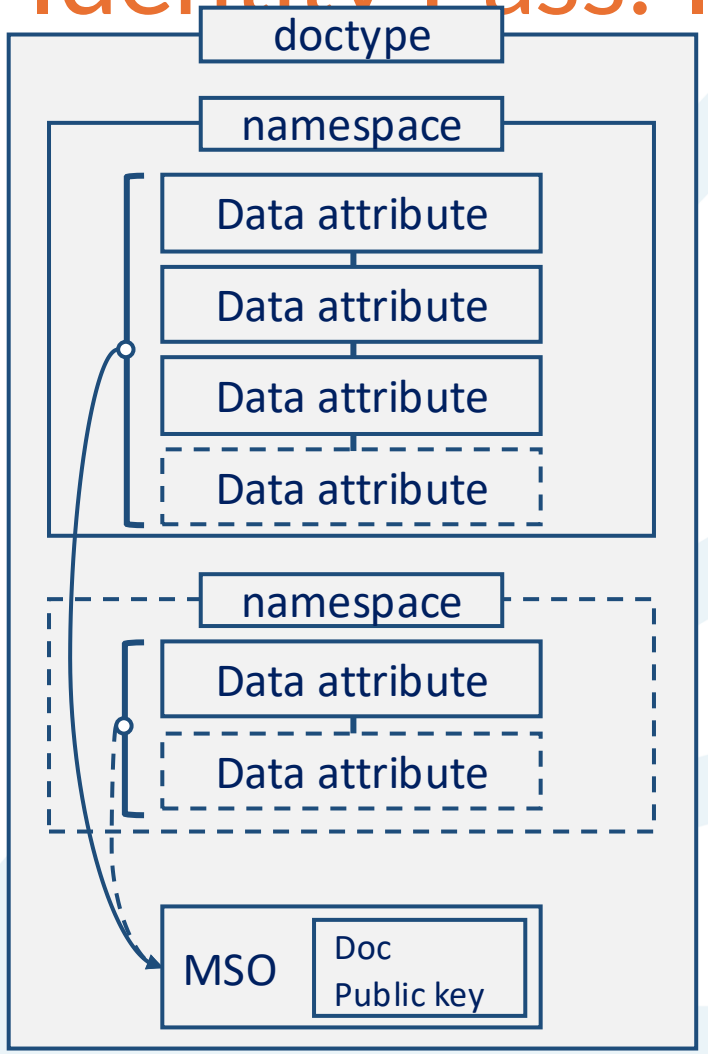
→ Our choice ISO/IEC 18013-5 (mDoc) for Wallet credentials



DC4EU → SD JWT

Image Source: https://www.linkedin.com/feed/update/urn:li:activity:7333379048112361472/

Identity Pass: ISO/IEC 18013-5 Data model



Reuse of directory data (common eduGAIN attributes):

- portrait (photo, *mandatory for identification documents*)
- givenNames
- sn (surname)
- email
- eduPersonAffiliation / eduPersonScopedAffiliation
- eduPersonEntitlement
- eduPersonAssurance
- schacHomeOrganization
- schacPersonalUniqueCode → European Student Identifier (ESI) - MyAcademicID
- schacDateOfBirth
- schacYearOfBirth
- schacPlaceOfBirth
- schacExpiryDate
- …

see also DC4EU education ID
(https://code.europa.eu/ebsi/json-schema/-/tree/main/schemas/vcdm1.1/education/verifiable-education-id)

Shibboleth

works like a Shibboleth authentication on-site → trained processes

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Institutional Use Cases

Part 3 · 11:50 – 12:00 · Morgan Persson

Which campus services does your institution want to unlock?

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Campus Cards are about Accessing Services

Identification
- **Identification**
- **Status verification**
- **Proof of entitlement**
- Single Sign On / 2FA / FIDO2
- Attendance check
  - *Check-in for exams*
  - *attendance at courses*
  - *Time recording*
- Electronic Signature (*of legal documents*)

Electronic payment / cashless campus
- Canteen & cafeterias
- Vending machines
- Printing / scanning (*secure & follow me printing*)
- Ticketing (*events and conferences*)

Physical Access Control
- Areas (*campus, parking lot*)
- Buildings
- Rooms
  - Classrooms
  - Labs
  - Computer rooms
  - Learning spaces
  - Offices
  - Accommodation facilities *(i.e., dorms)*
- Athletic facilities

Library services
- Access to / borrow
  - Physical media (book, audio and video media)
  - E-media (book, audio and video media)
- Learning spaces

Transport
- On campus services (*university shuttle service*)
- Public transport tickets & discounts

Discount and promotions
- **Discounts** on cultural activities
  - Museums
  - theaters
  - cinemas
- Shops
- Restaurants

On- & Off-Campus → On-Site Usage / Proximity Use-Cases + Online Services









eID.AS

Exercise: Pick One Use Case

In pairs or small groups · 7 minutes · select one institutional use case and answer:

1. Who is the service user?
2. Which status or entitlement must be verified?
3. Which institutional system holds the required information?
4. What should happen when the user taps?
5. Which information must – or must not – be disclosed?

Report back in one sentence: "At <institution>, our first eduTAP service is <service>, because <reason>."
One candidate per institution or group – we reuse it in the deployment block and at the reader.

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Designing an eduTAP Deployment

Part 4 · 12:00 – 12:30 · Alexander Loechel

Scenarios, roles, layers and integration points

Deployment Scenarios

All eduTAP components run as containers on an OCI-compatible container platform.

On-premises

- Operated inside the institution's own infrastructure
- Platform options: Docker Swarm, Kubernetes, K3s
- Direct control over infrastructure, operations and data
- Requires internal operational expertise and support capacity

Managed cloud

- Operated on a managed Kubernetes service
- Reduces the effort of running the underlying platform
- Decisions needed on hosting jurisdiction, data protection, connectivity, responsibilities
- Institutional integrations and IdM connections remain essential

Hybrid

- Reusable eduTAP components in a managed environment
- Institution-specific services and source systems stay on premises
- Useful where data residency and operational capacity differ per component
- Clear responsibility split is mandatory

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User and Operational Perspectives



Consumer / Tapper

- Obtains and stores passes in the smartphone wallet
- Presents a pass at a reader – no additional campus app
- Manages, renews and removes own passes



Support

- Assists users with provisioning and presentation problems
- Investigates failed taps, expired or revoked passes
- Manages operational incidents and escalations



Service Manager

- Configures pass classes, services and policies
- Owns integrations to source systems and entitlements
- Runs lifecycle processes: issue, update, revoke, expire

These are operational roles – not technical architecture layers.

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Functional and Technology Layers

Experience and Interaction	Tapper Portal · Pass Backoffice · Support interfaces · Reader / verifier applications
Institution-Specific Services	University REST services · service-specific business logic · adapters to local processes and systems
Reusable eduTAP Components	Pass creation, issuing and lifecycle · vendor-neutral pass APIs · wallet-provider integrations · callback and event processing · service discovery and reader libraries
Persistence and Events	Pass definitions and lifecycle state · operational data · event processing · audit and accounting
Institutional Systems	Student information system · identity management · library, access control, transport, payment, event systems
Authoritative Data Sources	Personal and organisational data · affiliations, roles, student status · entitlements · validity and lifecycle

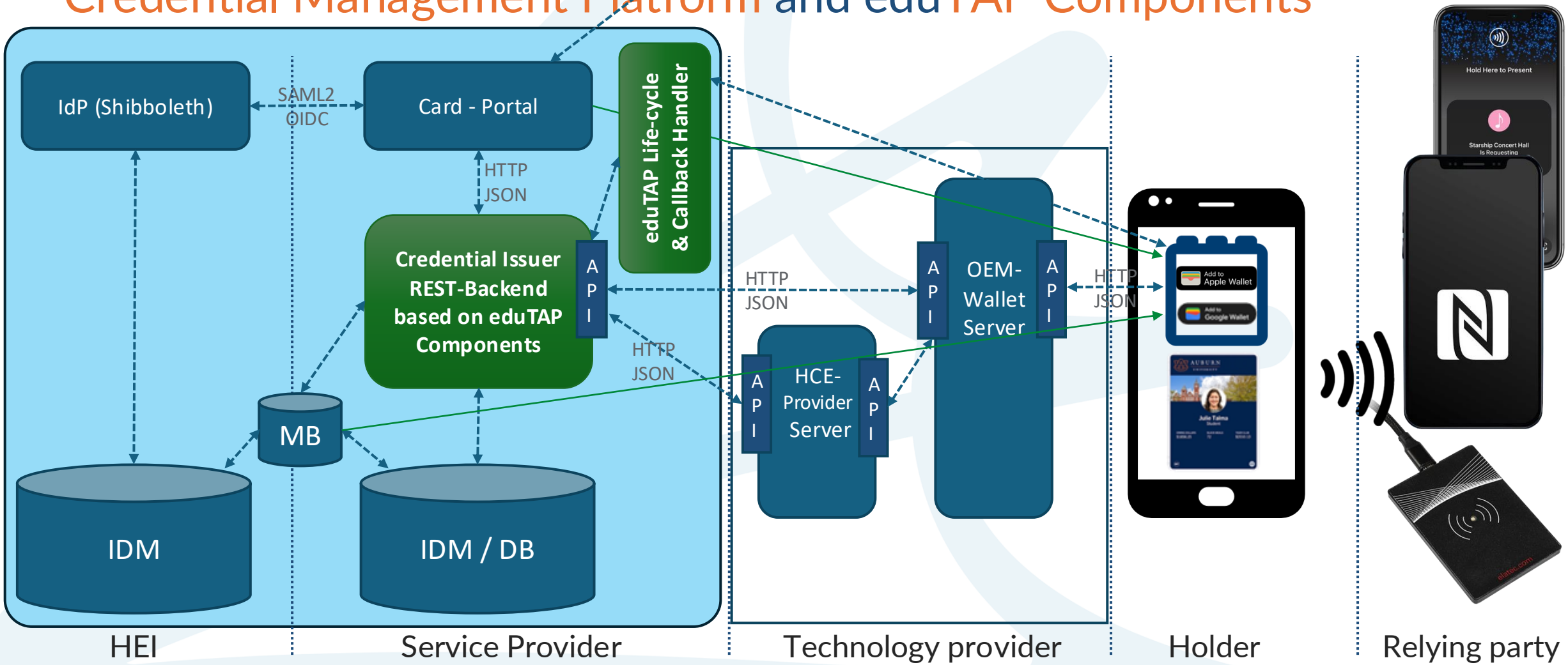
Trust, Authentication and Authorisation

- Institutional identity providers
- Shibboleth · SAML · OpenID Connect
- eduGAIN and related trust services
- Attribute and entitlement mapping
- Privacy, consent and selective disclosure

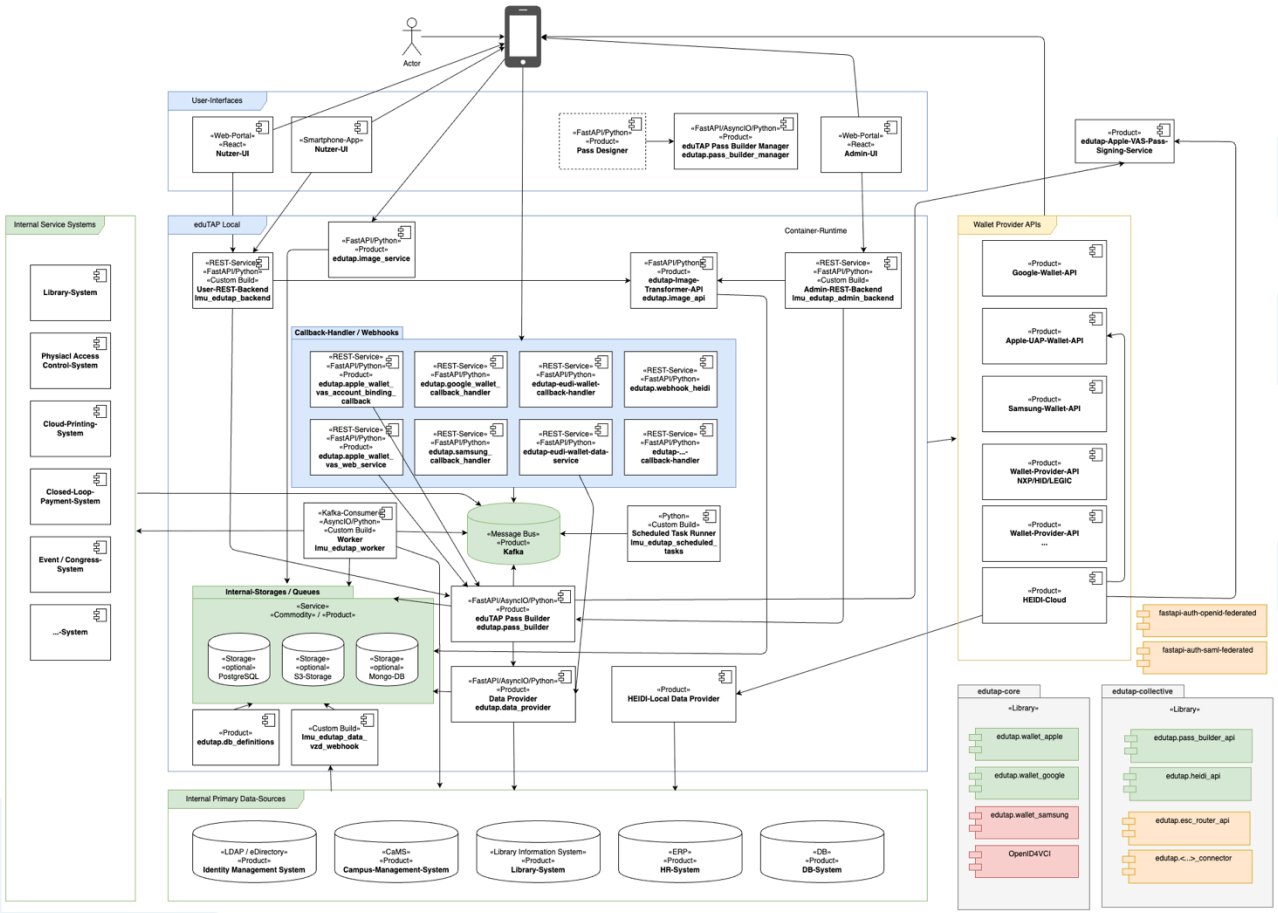
cross-cutting – affects all layers

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Credential Management Platform and eduTAP Components



Reference Deployment: eduTAP@LMU

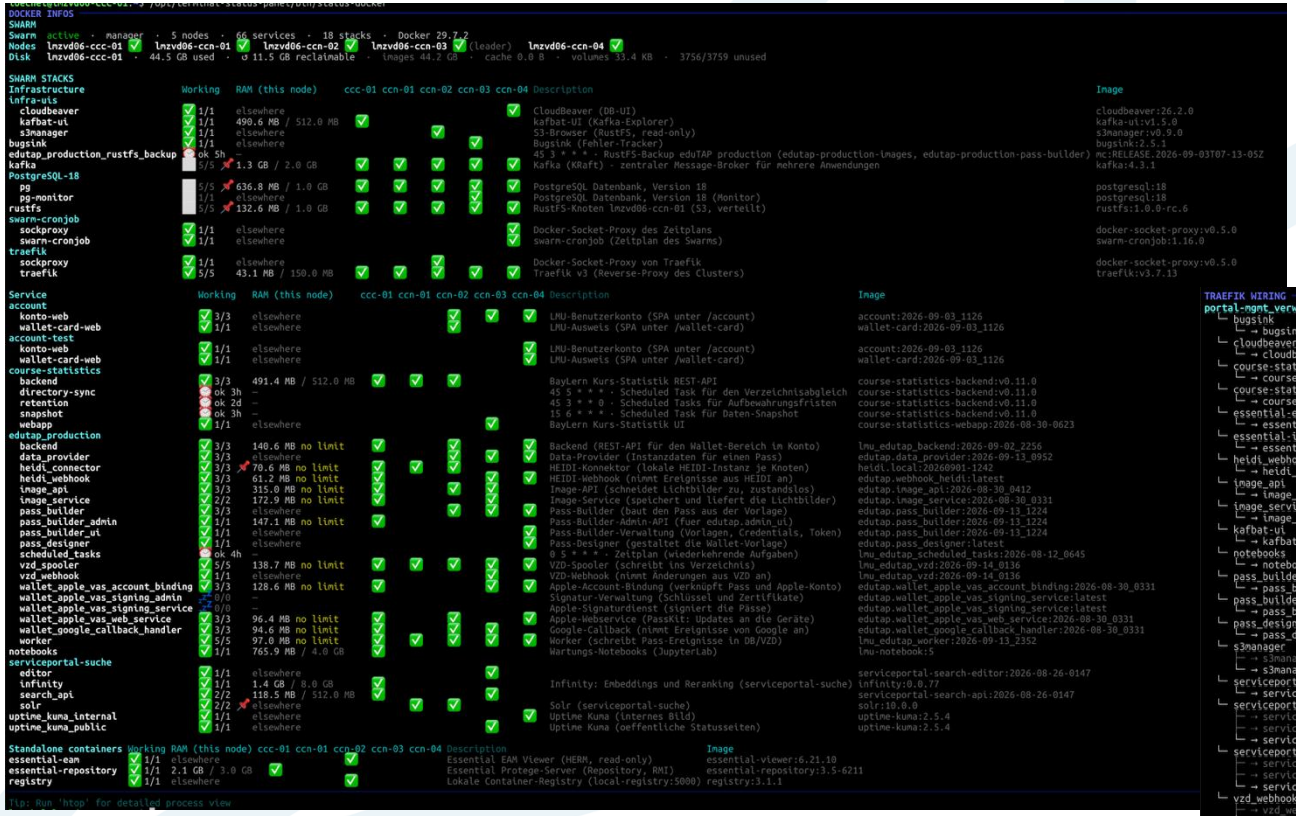


How the layers map to real components

- All components containerised – FastAPI / Python services on one container runtime
- User and Admin REST backends behind React portals (Nutzer-UI, Admin-UI, Pass Designer)
- One callback handler per wallet provider: Apple VAS, Google, Samsung, EUDI, HEIDI
- Kafka message bus, async worker and scheduled task runner drive the pass lifecycle
- Optional storages: PostgreSQL, S3, MongoDB
- Shared libraries: edutap-core (wallet_apple / google / samsung, OpenID4VCI) and edutap-collective (pass_builder_api, heidi_api, esc_router_api, connectors)
- Primary sources: IdM (LDAP / eDirectory), campus management, library, HR / ERP

Orientation map – we zoom into the relevant parts live; the full diagram is in the handout.

Deployment Status of eduTAP@LMU





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Deployment Discussion: Your Use Case

Take the use case you selected and identify …

- **Service owner –** who is accountable for running the service?
- **Authoritative data source –** where does the status or entitlement come from?
- **eduTAP components –** which reusable components are required?
- **Institutional integration –** which adapter or REST service must be built?
- **Credential / service pass –** identity pass, service pass, or both?
- **Reader / verification –** NFC, QR, or an existing terminal?
- **AAA requirements –** authentication, authorisation, and what must be recorded?

Output: a high-level institutional deployment blueprint.

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Pass Design

Part 5 · 12:30 – 12:40 · Alexander Loechel

How a service concept becomes a wallet pass

Designing a Service Pass

Pass class – decided once per service

- Identity pass versus service-specific pass – and how they coexist
- Visible information versus machine-readable information
- Branding and institutional design
- Target technology: Apple Wallet, Google Wallet, or other credential formats

Pass instance – decided per holder

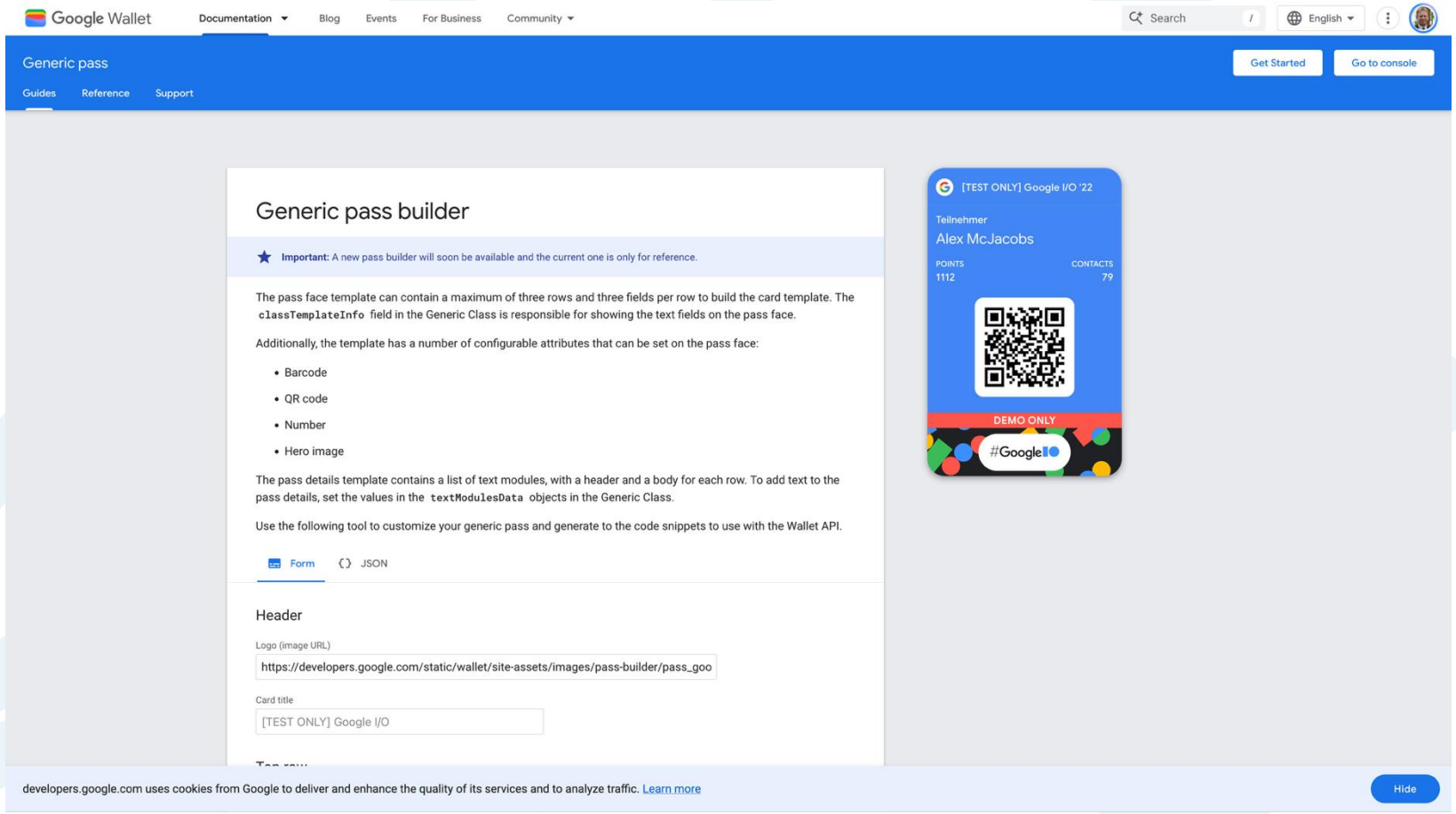
- Validity, expiration, revocation, and updates
- Which entitlement the pass actually proves
- Privacy and data minimisation – only what the service needs

Key question: what is the minimum information the pass must contain to enable the selected service?

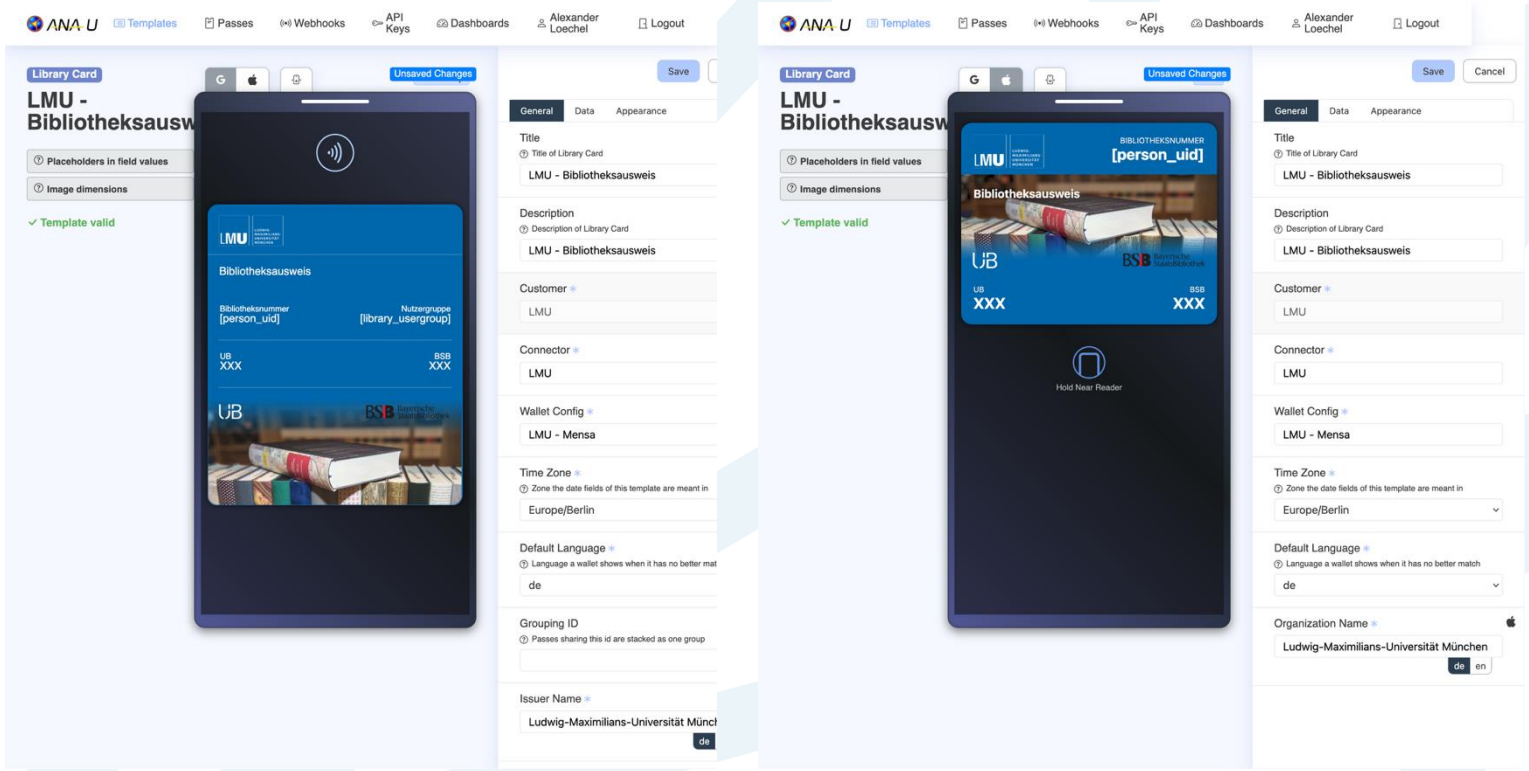
Apple Pass Designer



Google Pass Designer



Ana-U / Helix-Pass (HEIDI) Pass Designer



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Reader and Service Configuration

Part 6 · 12:40 – 12:50 · Morgan Persson

How the service provider consumes and validates a pass

Reader and Service Configuration

Presentation and validation

- The reader or verifier is the service provider's side of the tap
- Presentation: NFC (Apple VAS, Google Smart Tap), QR code, or barcode
- Validate in order: authenticity → validity → entitlement
- Online and offline operation – cached keys and revocation data
- Auto-presentment lets the wallet pick the right pass without user interaction

Integration and operations

- Connect the reader to the institutional service: door controller, library system, POS
- Logging and accounting – what has to be recorded, and for how long
- Handle expired, revoked, and unsupported passes explicitly
- Define fail-safe behaviour and the support escalation path
- Configure per service, not per person – entitlements come from the pass

Your service: what must the reader verify?

Work in pairs · 7 minutes · pick one service from your institution and answer the three questions.

1 Who is allowed in?
Authentication

- Which identity information does the service need – name, affiliation, student ID, photo, nothing but 'is a member'?
- How sure must you be: level of assurance, and how current must the data be?
- Does it have to work offline – and with what confidence?

2 What must be true right now?
Authorisation

- Which entitlement decides: role, enrolment status, booking, balance, age?
- Is it carried in the credential or looked up live in a backend?
- How quickly must a change or revocation take effect at the reader?

3 What has to be recorded?
Accounting

- What is the minimum you need to record – a count, a pseudonym, a name, a transaction?
- Who may see it, for what purpose, and how long is it kept (GDPR)?
- What happens when the reader says no – fallback, escalation, manual check?

Report back in one sentence: "For <service>, the reader verifies <credential>, checks <entitlement> and records <what>."
Your answers are the input for the reader configuration on the next slides.

End to End in One Tap

The same process for every service – library, door, canteen, shuttle, event.

1. Issue	2. Add	3. Present	4. Validate	5. Authorise	6. Account
Credential issuer builds the pass from authoritative data	Holder adds it via web, app, wallet or push provisioning	Holder taps the phone at the reader, or shows a code	Reader checks signature, validity and revocation	Service checks the entitlement carried in the pass	Access or transaction is recorded where required

Steps 1 and 2 happen once. Steps 3 to 6 happen at every tap – in well under a second.
Accounting is optional and service-specific: some services only need a yes or no, others need an auditable record.

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Conclusions, Next Steps and Questions

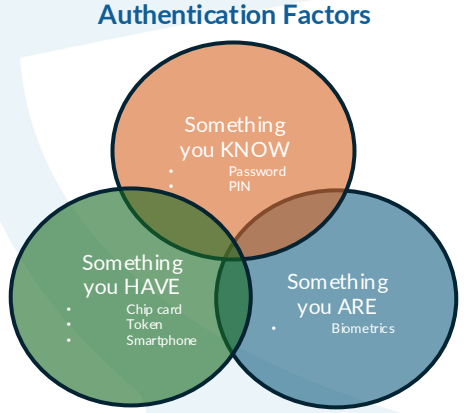
Part 7 · 12:50 – 13:00 · Alexander Loechel & Morgan Persson

From the guiding question to a first institutional pilot

Key Messages

1. **The European Student Card establishes the vision;** eduTAP turns it into operational service access.
2. **Identity / Status Verification alone is not enough: services need authentication, authorisation** and, where appropriate, ***accounting*.**
3. **Separate identity and service credentials:** a common identity pass can coexist with specialised service passes.
4. **Institutions keep control** over their services, data sources, entitlements and infrastructure.
5. **Reusable eduTAP components reduce effort;** institution-specific services connect them to local systems and processes.
6. **A deployment is not only a technical project:** it needs service ownership, support, governance, privacy decisions and lifecycle management.

- **Triple-A-System**
  - (Identification)
  - **Authentication**
  - **Authorization**
  - **Accounting**



Next Steps Towards a First Pilot

What you take home today

- One prioritised service-access use case
- An initial map of users, entitlements, data sources and systems
- A preliminary choice of deployment scenario
- An understanding of the pass and reader requirements
- A concrete first step towards an eduTAP pilot

Who you need on board

- Service owner, IT operations, identity management, data protection officer
- The provider of the target service: library, facility management, catering, transport
- Student representation – adoption is a user-experience question

Where to continue

- eduTAP.eu – documentation, components and community
- Reusable libraries: edutap-core and edutap-collective
- Talk to us:
  - Alexander Loechel (LMU Munich),
  - Morgan Persson (Lunds universitet)
  - Jose Filipe Alves (U.Porto)
  - Taha Youcef / Pierre Gabrielle (UPS)